Chapter 10. Trees

Staying Connected

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Trees: connected graphs that lose connectivity when any edge is removed.

10.1 Minimally connected graphs

Motivation. Four characterizations describe trees; Joyal's bijection counts labelled ones as n^{n-2}.

Walks contain paths [Supplement]. A shortest walk from x to y is a path: a repeated vertex would let us cut out a portion.

Theorem 10.1

For a connected simple graph G the following are equivalent:
(1) G is **minimally connected**: deleting any edge disconnects it;
(2) G contains no cycle.

Definition 10.2 (Tree)

A connected simple graph satisfying (1), equivalently (2), is a **tree**.

Proof of Theorem 10.1.

Step (1)⇒(2). If a cycle C contains ab, a path using ab reroutes along $C-ab$, so $G-ab$ is connected, contradicting (1).

Step (2)⇒(1). If $G-AB$ is connected, a B–A path P avoids AB, and $AB \cup P$ is a cycle. ■

Corollary 10.3

A connected graph is a tree iff every two vertices are joined by exactly one path.

Proof.

Step (unique paths ⇒ tree). If $G-rs$ were connected, r, s would be joined by rs and by a path avoiding it.

Step (tree ⇒ unique paths) [Supplement]. Let $P \neq Q$ be x–y paths, u the last vertex up to which they agree, and w the first vertex after u on P lying on Q after u (y qualifies). The u–w segments of P and Q meet only at u, w and form a cycle, contradicting Theorem 10.1. ■

Theorem 10.4

Every tree on n vertices has $n-1$ edges. Conversely, every connected graph on n vertices with $n-1$ edges is a tree.

Lemma 10.5

A tree on $n \geq 2$ vertices has at least two vertices of degree 1 (**leaves**).

Proof. Take a longest path p. Another neighbour b of an endpoint a would either extend p ($b \notin p$) or close a cycle with ab ($b \in p$). So both endpoints are leaves. ■

Proof of Theorem 10.4.

Step 1 (trees). Induction on n; $n = 1$ is trivial. A tree T on $n+1$ vertices has a leaf l (Lemma 10.5); $T-l$ is a tree on n vertices with $n-1$ edges, so T has n.

Step 2 (converse) [Supplement]. Let G be connected, n vertices, $n-1$ edges. Deleting edges of cycles preserves connectivity and ends with a tree on n vertices, which has $n-1$ edges by Step 1. So nothing was deleted and G is a tree. ■

A **forest** is a graph whose components are trees.

Proposition 10.6

A forest on n vertices with k components has $n-k$ edges.

Proof. By Theorem 10.4 each component has one more vertex than edges. ■

Counting labelled trees. Trees on $[n]$ number $1, 1, 3, 16$ for $n = 1, \dots, 4$, suggesting n^{n-2} (unlabelled trees: book, Chapter 18).

Theorem 10.7 (Cayley's formula)

There are $A_n = n^{n-2}$ trees with vertex set $[n]$.

Proof (Joyal).

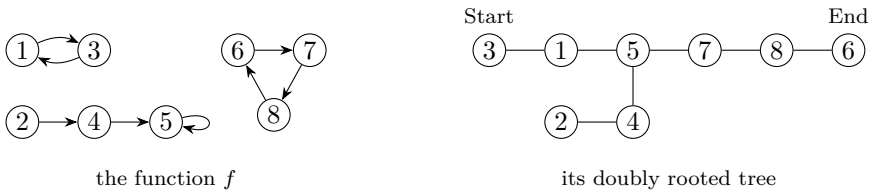
Step 1 (doubly rooted trees). Marking a Start and an End vertex (possibly equal) gives $n^2 A_n$ *doubly rooted trees*. We biject the n^n functions $f : [n] \to [n]$ onto them, so $n^2 A_n = n^n$.

Step 2 (functions to trees). Let $C = \{c_1 < \cdots < c_k\}$ be the elements on cycles of f and $d_i := f(c_i)$. Put $d_1, \dots, d_k$ along a path (Start d_1, End d_k) and add $\{j, f(j)\}$ for $j \notin C$. Iterating f reaches C, so this is connected, with $(k-1) + (n-k) = n-1$ edges: a tree (Theorem 10.4).

Step 3 (inverse). For j off the Start–End path, $f(j)$ is the neighbour of j towards the path. On the path read $d_1, \dots, d_k$ from Start to End, sort them as $c_1 < \cdots < c_k$ and set $f(c_i) := d_i$. This recovers f uniquely. ■

Example 10.8

$n = 8$: $f(1) = 3$, $f(2) = 4$, $f(3) = 1$, $f(4) = 5$, $f(5) = 5$, $f(6) = 7$, $f(7) = 8$, $f(8) = 6$. Cycles $(13), (5), (678)$, $C = \{1, 3, 5, 6, 7, 8\}$, $(d_i) = (3, 1, 5, 7, 8, 6)$: the Start–End path is $3 - 1 - 5 - 7 - 8 - 6$, and $2 - 4 - 5$ hangs off it.



the function f its doubly rooted tree

Figure. *Joyal's bijection for Example 10.8: the cyclic part becomes the Start–End path; other elements hang off along $j \mapsto f(j)$.*

A **rooted tree** has one marked root (n^{n-1} on $[n]$); a **rooted forest** is a forest of rooted trees.

Corollary 10.9

There are $(n+1)^{n-1}$ rooted forests on $[n]$.

Proof. Join every root to a new vertex $n+1$; conversely delete $n+1$ and make its neighbours roots. These are inverse bijections with the $(n+1)^{n-1}$ trees on $[n+1]$ (Theorem 10.7). ■

10.2 Minimum-weight spanning trees. Kruskal's greedy algorithm

Motivation. Greedy is optimal for spanning trees because forests have an exchange property.

A **spanning tree** of connected G is a tree on $V(G)$ using edges of G; one exists (delete cycle edges).

Example 10.10 (Minimum-weight spanning tree)

Given a connected simple graph G and weights $w : E(G) \to \mathbb{R}_+$, find a spanning tree T minimising $w(T) = \sum_{e \in T} w(e)$. (For $n = 20$, $20^{18} > 2.5 \cdot 10^{23}$ trees: over 91 years at 10^9 per second.)

Kruskal's greedy algorithm. Start with $T = \emptyset$; at each step add a minimum-weight edge creating no cycle. T stays a forest; with fewer than $n-1$ edges it is disconnected and some edge of G joins two components, so after $n-1$ steps T is a spanning tree.

Greedy can fail elsewhere: for two disjoint edges of minimum total weight in a path $2, 1, 2$ plus a disjoint edge 10, greedy gives 11, not 4 (Chapter 11).

Lemma 10.11 (Exchange property)

If F, F' are forests on the same vertex set and F has fewer edges than F', then some edge e of F' can be added to F so that $F \cup e$ is still a forest.

Proof. Otherwise every edge of F' lies within a component of F, so F' has at least as many components as F and, by Proposition 10.6, at most as many edges, a contradiction. ∎

Theorem 10.12

Kruskal's greedy algorithm finds a minimum-weight spanning tree.

Proof.

Step 1. Let $T = \{t_1, \dots, t_{n-1}\}$ be the greedy tree and $H = \{h_1, \dots, h_{n-1}\}$ a lighter spanning tree, both by increasing weight ($w(t_i)$ increase since eligible edges only shrink).

Step 2. Let i be minimal with $\sum_{j \le i} w(h_j) < \sum_{j \le i} w(t_j)$; minimality gives $w(h_i) < w(t_i)$.

Step 3. By Lemma 10.11 for $T_{i-1} = \{t_1, \dots, t_{i-1}\}$ and $H_i = \{h_1, \dots, h_i\}$, some h_j, $j \le i$, can be added to T_{i-1}; as $w(h_j) \le w(h_i) < w(t_i)$, greedy would not pick t_i, a contradiction. ∎

10.3 Adjacency matrices of graphs

Motivation. Matrix powers count walks.

Definition 10.13 (Adjacency matrix)

For a graph G on labelled vertices $1, \dots, n$, the **adjacency matrix** $A = A_G$ has A_{ij} equal to the number of edges between i and j (from i to j for digraphs). It is symmetric for undirected graphs.

Example 10.14

For the undirected graph on A, B, C, D with edges AB, AC, AD, BD:

$$A_G = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

Example 10.15

For the digraph with edges $A \to B$, $A \to C$, $B \to D$, $D \to A$:

$$A_H = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Theorem 10.16

$(A^k)_{ij}$ is the number of walks of length k from i to j.

Proof. Induction on k; $k = 1$ is the definition. With $B = A^k$ counting length-k walks, walks of length $k+1$ from i to j with next-to-last vertex z number $b_{iz}a_{zj}$; summing over z gives $(A^{k+1})_{ij}$. ∎

Theorem 10.17

A simple graph G on n vertices is connected iff all entries of $(I + A)^{n-1}$ are positive.

Proof. Paths have length $\le n-1$, so G is connected iff for all $i \neq j$ some $(A^k)_{ij} > 0$, $1 \le k \le n-1$. Now $(I+A)^{n-1} = \sum_{k=0}^{n-1} \binom{n-1}{k} A^k$ with positive coefficients, nonnegative terms, and positive diagonal from I. ∎

10.4 The number of spanning trees of a graph

Motivation. Kirchhoff: spanning trees are counted by a determinant, via Cauchy–Binet.

A **spanning tree** of a digraph is a subgraph whose underlying undirected graph is a spanning tree.

Definition 10.18 (Incidence matrix)

For a loopless digraph with vertices $v_1, \dots, v_n$ and edges $e_1, \dots, e_m$, the **incidence matrix** A is $n \times m$ with $a_{ij} = 1$ if v_i is the start of e_j, -1 if v_i is its end, and 0 otherwise.

Theorem 10.19

Let A be the incidence matrix of a loopless digraph G and A_0 the matrix obtained by deleting any one row. Then the number of spanning trees of G is $\det(A_0 A_0^{\top})$.

Proof. Delete the last row. If $m < n - 1$, G is disconnected and both sides are 0.

Step 1 (claim). For every $(n-1)\times(n-1)$ submatrix B of A_0 (edges forming G'), $|\det B| = 1$ if G' is a spanning tree and $\det B = 0$ otherwise. Induction on n; for $n = 2$, $B = (\pm 1)$.

Step 2 (a leaf other than v_n). If v_i, $i \neq n$, has degree 1 in G', its row has a single entry ± 1; expanding reduces to $G - v_i$, $G' - v_i$, and G' is a spanning tree iff $G' - v_i$ is.

Step 3 (no such leaf). Then G' is not a spanning tree (Lemma 10.5), and some vertex has degree 0 (otherwise $\geq \frac{2(n-1)+1}{2} > n-1$ edges). If it is v_i, $i \neq n$, B has a zero row; if it is v_n, each column has one 1 and one -1, so the rows sum to 0. Either way $\det B = 0$.

Step 4 (Cauchy–Binet). $\det(A_0 A_0^\top) = \sum_B (\det B)^2$ over all $(n-1)\times(n-1)$ submatrices B, which by Step 1 is the number of spanning trees. ■

Theorem 10.20 (Matrix-tree theorem)

Let U be a simple undirected graph with vertices $v_1, \ldots, v_n$ and L_0 the $(n-1)\times(n-1)$ matrix with $\ell_{ii} = \deg v_i$, $\ell_{ij} = -1$ if $v_i v_j$ is an edge, and 0 otherwise $(1 \leq i, j \leq n-1)$. Then U has exactly $\det L_0$ spanning trees.

Proof.

Step 1. Replace each edge by two opposite directed edges (G). Then $(A_0 A_0^\top)_{ii} = 2\deg_U v_i$ and $(A_0 A_0^\top)_{ij} = -2$ for adjacent v_i, v_j, so $A_0 A_0^\top = 2L_0$ and $\det(A_0 A_0^\top) = 2^{n-1}\det L_0$.

Step 2. Each spanning tree of U gives exactly 2^{n-1} spanning trees of G (orient each edge), and all arise so. By Theorem 10.19, $2^{n-1}\tau(U) = 2^{n-1}\det L_0$. ■

Example 10.21

K_n: L_0 has $n-1$ on the diagonal, -1 elsewhere. Adding all rows to the first gives $(1, \ldots, 1)$; adding it to the others gives a triangular matrix with diagonal $1, n, \ldots, n$, so $\det L_0 = n^{n-2}$.

Example 10.22

The **complete bipartite graph** $K_{m,n}$ (parts of sizes m, n, all mn edges between) has $n^{m-1}m^{n-1}$ spanning trees. Delete a vertex of the n-part: L_0 has diagonal blocks nI_m, mI_{n-1} and -1 between parts. Adding all rows to the first gives $(1, \ldots, 1, 0, \ldots, 0)$; adding it to the last $n-1$ rows makes it block triangular with determinant $n^{m-1}m^{n-1}$.

Theorem 10.23 (Matrix-tree theorem, eigenvalue version)

Let L be the $n \times n$ **Laplacian** of U (defined like L_0 but over all n vertices), with eigenvalues $\lambda_1, \ldots, \lambda_{n-1}, \lambda_n = 0$. Then the number of spanning trees of U is $\frac{1}{n}\lambda_1\lambda_2\cdots\lambda_{n-1}$.

Proof [Supplement]. Rows of L sum to 0, so $\lambda_n = 0$ and $\det(xI - L) = x\prod_{i<n}(x - \lambda_i)$ has x-coefficient $(-1)^{n-1}\lambda_1\cdots\lambda_{n-1}$. Expanding, this coefficient is also $(-1)^{n-1}$ times the sum of the n principal $(n-1)\times(n-1)$ minors, each equal to $\tau(U)$ by Theorem 10.20 (relabel with v_k last). So $\lambda_1\cdots\lambda_{n-1} = n\,\tau(U)$. ■

If U is d-**regular**, $L = dI - A$, so L has eigenvalues $d - \alpha_i$.

Example 10.24

K_n: $A + I = J$ has eigenvalues $n, 0, \ldots, 0$, so A has $n-1, -1, \ldots, -1$ and $L = (n-1)I - A$ has $0, n, \ldots, n$; Theorem 10.23 gives $\frac{1}{n}n^{n-1} = n^{n-2}$.

10.5 Selected exercises

Motivation. Tree characterizations at work: degree sequences, complements, and the cut property.

Exercise 10.1 (Degree sequences of trees)

Let $n \geq 2$ and let $a_1 \geq a_2 \geq \cdots \geq a_n$ be positive integers with $a_1 + \cdots + a_n = 2n - 2$. Prove that there is a tree on n vertices whose vertices have degrees $a_1, \ldots, a_n$.

Solution. Induction on n, for any order of the a_i.

Step 1 (initial step). $n = 2$: $a_1 = a_2 = 1$, a single edge.

Step 2 (a 1 and a part ≥ 2). Let $n \geq 3$. All $a_i \geq 2$ would give sum $\geq 2n$, so say $a_n = 1$; all $a_i = 1$ would give $n < 2n - 2$, so say $a_1 \geq 2$.

Step 3 (reduction). $a_1 - 1, a_2, \ldots, a_{n-1}$ are positive with sum $2(n-1) - 2$, so by induction a tree T' on $v_1, \ldots, v_{n-1}$ has these degrees.

Step 4 (adding a leaf). Add v_n and $v_1 v_n$: connected, n vertices, $n-1$ edges, so a tree (Theorem 10.4) with degrees $a_1, \ldots, a_n$. ■

Exercise 10.10 (A graph or its complement is connected)

Prove that for every simple graph G on $[n]$, at least one of G and its complement $\bar{G}$ is connected. Give an example where both are connected.

Solution. Let G be disconnected and $x \neq y$.

Step 1 (different components). Then $xy \in E(\bar{G})$.

Step 2 (same component). Take z in another component; $x\,z\,y$ is a path in $\bar{G}$.

So $\bar{G}$ is connected. Example: the path $1\,2\,3\,4$ has complement $13, 14, 24$, the path $3\,1\,4\,2$. ■

Exercise 10.24 (The cut property)

A *cut* in a connected graph G is a minimal set S of edges whose removal splits G into two components. Let w be a weight function on the edges, let S be a cut, and let $e \in S$ satisfy $w(e) < w(f)$ for all other $f \in S$. Prove that every minimum-weight spanning tree of G contains e.

Solution.

Step 1 (the shape of a cut). Let V_1, V_2 be the vertex sets of the components of $G - S$. Edges between V_1, V_2 lie in S. Conversely, if $f \in S$ had both ends in one V_i, then $S \setminus \{f\}$ would still split G into two components, contradicting minimality. So S is exactly the set of V_1–V_2 edges.

Step 2 (exchange). Let T be a minimum-weight spanning tree with $e = uv \notin T$, $u \in V_1$, $v \in V_2$. The unique u–v path in T (Corollary 10.3) has an edge f from V_1 to V_2; $f \in S$, $f \neq e$, so $w(f) > w(e)$.

Step 3 (a cheaper tree). $T - f$ has $n-2$ edges, hence two components (Proposition 10.6), separating u, v (else a second u–v path). So $T' = T - f + e$ is connected with $n - 1$ edges, a spanning tree (Theorem 10.4), and

$$w(T') = w(T) - w(f) + w(e) < w(T),$$

a contradiction. Hence $e \in T$. ■